

Miguel Francisco Ruales Pianda

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SUMMARY

Software Engineering student with 2+ years of full-stack experience and recent deep focus on agentic AI systems. Built and led model-agnostic multi-agent workflows, RAG pipelines, and long-form report generation systems with evaluation, prompt versioning, tracing, and human-in-the-loop review. Strong background in architecture decisions, technical leadership, and shipping production software for U.S. clients.

EXPERIENCE

Full-Stack / AI Engineer

[Project Zena](#) (Brandtuit AI) - [Melt Studio](#) | Oct 2025 - Present

- Designed and maintained AI workflows for generating, modifying, and exporting brand-management reports in DOCX and PPT formats, including AI-generated visual aids.
- Defined agent and graph patterns in LangGraph, including advanced orchestration patterns such as durable execution, dynamic routing, and state-aware processing.
- Led research and implementation of context-engineering techniques, document indexing strategies, and retrieval improvements for long-form report consistency.
- Improved quality of research insights and long-document coherence through prompt iteration, structured outputs, and HITL review by brand-management professionals.
- Guided three teammates, performed AI-related code reviews, created internal best practices, and trained the team on LangChain updates and implementation patterns.
- Worked directly with the client on prioritization and technical decisions, and supported product delivery across a beta program used by three U.S. companies.
- Led the architecture of a model-agnostic multi-agent system with roughly 20 agents for research, planning, report editing, and AI-assisted export generation.
- Implemented RAG and context-engineering workflows using PostgreSQL vector search and Docling for document ingestion and retrieval.
- Built prompt middleware, dynamic tools, structured outputs, evals, prompt versioning, and Langfuse tracing to improve consistency and observability.
- Reduced costs by about 50% and runtime to about one-third while improving long-form coherence and client acceptance of the generated reports.

Full-Stack Web Engineer

[Caramelo Escaso](#) - [Melt Studio](#) | Sep 2025 - Oct 2025

- Led the redesign and modernization of an organizational-climate measurement application, improving usability, performance, and maintainability.
- Built the frontend with React, Next.js, shadcn/ui, and Tailwind CSS using pixel-perfect Figma specifications.
- Added backend features with TypeScript, tRPC, Drizzle, and PostgreSQL, and reduced coupling through cleaner Context API patterns.
- Worked directly with design and clients to align priorities around user experience and data security.
- Delivered the full redesign in one month while significantly reducing technical debt.

Full-Stack Developer & Product Founder / PM

[Hotel Management Platform](#) | Pasto, Colombia | Nov 2024 - Jun 2025

- Built a complete hotel management platform from scratch, covering backend, frontend, automation, analytics, and authentication.
- Developed 100+ REST endpoints in NestJS, including SSE flows and scheduled cron jobs.

- Integrated government APIs and automated legacy government systems using Playwright scraping.
- Designed JWT-based authentication with custom decorators and guards, plus protected routes on the Next.js frontend.
- Modeled and optimized PostgreSQL with Prisma, wrote tests in Jest using TDD, and documented endpoints with Swagger.

Team Lead

Hackathon Parquesoft 2024 | Pasto, Colombia | Nov 2024

- Led a four-person team as PM and Scrum Master to first place with a regional technology MVP.
- Coordinated daily sprints, delegated tasks, and delivered a working MVP in less than one week.
- Designed wireframes, database models, and the technology stack, and deployed a React Native + Expo app using Firebase.

Private Programming Tutor

Software Engineering Tutor | Pasto, Colombia | 2024 - 2025

- Delivered group tutorials on OOP and Python to classes of 20 students.
- Provided one-on-one support that improved classmates' project performance.
- Conducted basic programming and VBA automation training for a Bancolombia manager.

EDUCATION

Universidad Cooperativa de Colombia — B.Sc. in Software Engineering, 4th semester, GPA 4.8/5.0

Universidad Pontificia Bolivariana — Web Applications Bootcamp (800 hours)

Departmental Certification in Digital Ecosystem Innovation Leadership — 2024 - Present

SKILLS

Languages: TypeScript, JavaScript, Python

Frontend: Next.js, React, React Native, Tailwind CSS, shadcn/ui, React Query

Backend: NestJS, Node.js, Express, FastAPI, tRPC, Celery

AI / Data: LangChain, LangGraph, Docling, Chroma, LangSmith, PostgreSQL vector search

Databases / DevOps: PostgreSQL, Prisma, Drizzle, Firebase, Docker, GitHub Actions, AWS

Tools / Practices: JWT, Swagger, Playwright, Pydantic, Jest, TDD

LANGUAGES

Spanish: Native

English: Fluent / C2